EXP No: 4 VERIFICATION OF THEVENIN'S THEOREM Pg.No

DATE: 24/2/26

AIM:

To verify thevenin's theorem practically and theoretically for the given DC circuit.

APPARATUS REQUIRED:

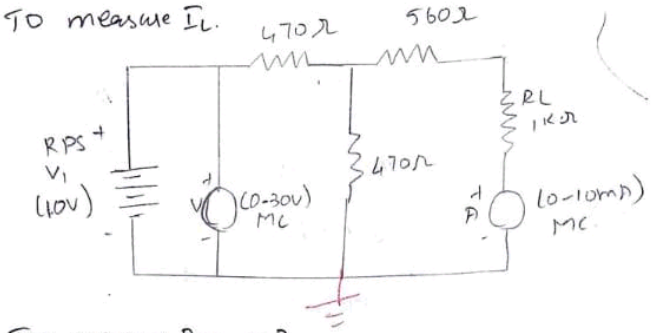
Sl.NO	APPARATUS	SPECIFICATION	QUANTITY
1	RPS	(0-30V)	1
2	Voltmeter	(0-30V) MC	1
3	Ammeter	(0-10mA) MC	1
4	Resistor	470Ω, 560Ω, 1KΩ	2, 1, 1
5	Bread Board	-	1
6	Multimeter	-	1

PROCEDURE:

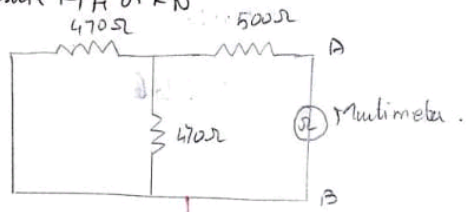
1) Make the connections as per the circuit Diagram.
2) Vary the RPS and set an Input voltage of 10V.
3) Note down the voltmeter reading (V_i) and ammeter reading (I_L) in Tabular Column.
4) Switch off the supply and make connections for circuit Diagram: 2.
5) Measure the Thevenin's resistance R_{TH} = Norton's resistance R_N.
6) Switch off the supply and make connections for circuit Diagram 3.
7) Set an input voltage of 10V in the RPS & note down the voltmeter readings V_i and V_{TH} (= V_{OC}) in Tabular Column: 3.

CIRCUIT DIAGRAM:

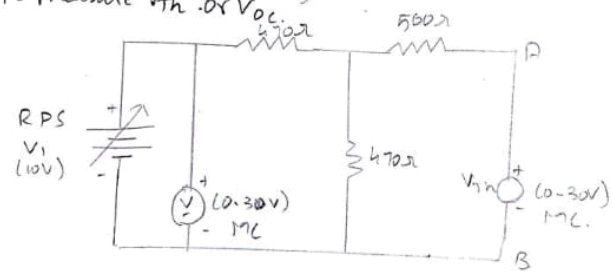
To measure I_L.



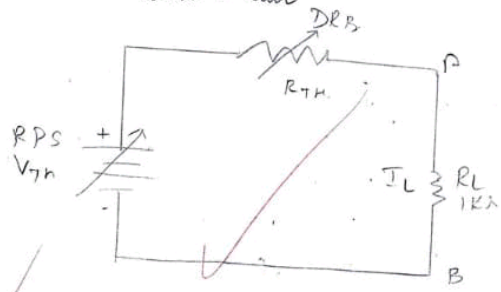
To measure R_{TH} or R_N



To measure V_{Th} or V_{OC}.



Thevenin's equivalent circuit



TABULAR COLUMN: 1

To measure I_L

V_i (volts)	I_L (amps)
10V	2.81 mA

TABULAR COLUMN: 3

To measure V_{Th} or V_{OV}

V_i (volts)	V_{Th} (volts)
10V	5V

TABULAR COLUMN: 2.

To measure R_{Th} or R_N

From circuit diagram 2, $R_{Th} = R_N = 79\,k\Omega$

MODEL CALCULATION:

Practical value of I_L (from tabulation) $= 2.3$ mA

Verification of Thevenin's theorem.

$$I_L = \frac{V_{Th}}{(R_{Th}+R_L)} \Rightarrow I_L = 2.22\,mA$$

Theoretical Calculation of I_L, $R_{Th}(R_N)$

V_{TH} for the given circuit

By voltage division rule

$$V_{Th} = \frac{V_S \times R_S}{R_1 + R_2}$$

$$V_{TH} = \frac{10V \times 470\,\Omega}{(470+470)\,\Omega}$$

$$\boxed{V_{TH} = 5V}$$

(No current flow through R_2 when R_2 is open circuit)

R_{TH}: R_1 & R_2 is parallel when voltage source is short circuited

$$R_{TH} = \frac{470 \times 470}{470+470} + 560$$

$$= 235 + 560$$

$$= 795\,\Omega$$

$$R_{TH} = 795\,\Omega$$

Find I_L:

R_{Th} in series with R_L

$\therefore R_{Eq} = R_L + R_{Th} \Rightarrow 1.795\,k\Omega$

$$I_L = \frac{V_{Th}}{R_{eq}} = \frac{5V}{1.795 \times 10^3\,\Omega}$$

$$\boxed{I_L = 2.79\,mA}$$

8) Switch off the supply and make connections for circuit Diagram 4.

9) Set an input of 10V in the RPS and note down the voltmeter reading V_i and ammeter reading $I_N = (I_{SC})$ in tabular column 2.

10) Calculate the I_L value using the formula Thevenin's Theorem.

$$I_L = V_{TH} / (R_{TH} + R_L)$$

11) Norton's theorem.

$$I_L = I_N * R_N / (R_N + R_L)$$

12) Theoretically verify the Norton's theorem.

Mark split-up

S.NO	Description	Marks Alloted	Marks obtained
1	Experimental setup / Connections	20	20
2	Observation / Reading	20	20
3	Calculations / Results	20	20
4	Viva	20	20
5	Record	20	20
		Total	90

90

17/5/24

Result:

Thus Thevenin's theorem is verified practically and theoretically.